

Preference-Graph Data Quality Pipeline and Evaluation Framework

1 Multi-Ranker Collection

For each query, collect top-N rankings from diverse retrievers and LLM rankers.

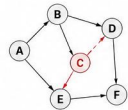


...
Up to 8 rankers

Output: Ranked lists per query from multiple models and configurations.

2 Preference Graph Construction

Aggregate pairwise preferences into a weighted directed graph.



Edge $u \rightarrow v$ indicates that more rankers prefer u over v .

Output: Weighted directed graph $G = (V, E, w)$.

3 Inconsistency Measurement

Quantify structural data quality (DQ) problems in the graph.

- Cyclicity (cycle ratio)
- BEW (Broken Edge Weight)
- PIC (Pairwise Inconsistency Count)
- SCC size / count

Example Cycle



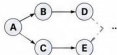
Output: DQ metrics (pre-repair).

4 Graph Repair (FAS)

Apply Feedback Arc Set (FAS) solvers to break cycles and obtain a DAG.

Exact FAS Solver
(optimal)

Greedy FAS Solver
(approximate)



Output: Repaired DAG G' (edges in FAS removed).

5 Retrieval Ranking & Evaluation

Define ranking scores on the (repaired) graph and evaluate retrieval quality.

Scoring methods:

- Hybrid FAS (our method)
- CombSUM (baseline)
- RRF (baseline)



Metrics:

nDCG@5/10/20, Recall@5/10/20, MRR@10

Output: Effectiveness scores per query $\times$ method $\times$ regime.

6 Analysis & Diagnosis

Analyze when (and why) repair helps or fails.



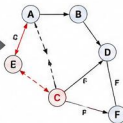
Output: Data quality insights, failure taxonomy, and practical guidance.

Concrete Example: One Query (Illustrative)

Rankers' Top Lists (k=3)

BM25: A > B > C > D > E > F
TF-IDF: B > A > D > C > F > E
MiniLM: A > D > B > C > E > F
LLM: D > B > A > C > E > F
...
(Up to 8 rankers)

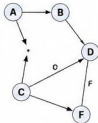
Preference Graph (Pre-Repair)



DQ Metrics (Pre)	
Cyclicity	0.28
BEW (sum w)	31
PIC	18
SCC (largest)	4

FAS Repair

Repaired Graph (Exact FAS)



DQ Metrics (Post)	
Cyclicity	0.00
BEW (sum w)	0
PIC	0
SCC (largest)	1

Retrieval Quality (nDCG@10)

Method	Pre-Repair	Post-Repair	Δ
Hybrid FAS	0.462	0.466	+0.004
CombSUM	0.513	0.513	0.000
RRF	0.528	0.528	0.000

Diagnosis (This Query)

Baselines stronger (already well aligned)

Counts toward failure taxonomy.

Repair improves DQ substantially, but retrieval quality **does not** improve here.



Evaluation Scale

1,020 queries $\times$ 4 extraction regimes
 $\times$ 3 datasets = 32 experiments
366 methods evaluated



Extraction Regimes

- Range
- Majority
- Public Vote
- Public Vote (top-4)



FAS Solvers

- Exact (optimal)
- Greedy (approximate)



Baselines

- CombSUM
- RRF



Datasets

MS MARCO Dev
TREC DL 2019
HotpotQA (DPR top-100)



Open Resources (CARB)

Consistency-Aware Ranking
Benchmark + full artifacts
(see supplementary material)